

Case Study Report: Vehicle Fuel Efficiency Optimizer

Using Neural Networks for Driving Behavior Optimization

Executive Summary

This case study explores how neural networks can be used to analyze driving behavior and improve vehicle fuel efficiency. Using real telemetry data from Max Verstappen's 2024 Miami Grand Prix, a feedforward neural network was developed to predict a fuel efficiency score based on speed, gear selection, and throttle position. The model learns non-linear relationships between these driving parameters and fuel usage, allowing identification of more fuel-efficient driving patterns.

Project Highlights

- Dataset: 20,656 Formula 1 telemetry samples
- Input features: Speed, Gear, Throttle
- Model: Feedforward neural network (3 → 10 → 8 → 1)
- Task type: Regression

1. Problem Description

Fuel efficiency is an important factor in both automotive engineering and environmental sustainability. While vehicle design plays a major role, driver behavior such as acceleration, gear changes, and throttle usage also significantly affects fuel consumption.

The objective of this project is to build a neural network that predicts fuel efficiency based on key driving behavior variables:

- Vehicle speed (km/h)
- Gear selection
- Throttle position (%)

By learning these relationships, the model can help identify driving patterns that reduce fuel consumption without sacrificing performance.

2. Dataset Description

The dataset used in this project comes from Formula 1 telemetry data recorded during the 2024 Miami Grand Prix. It contains high-frequency measurements captured throughout the race.

Dataset Details

- Driver: Max Verstappen
- Total samples: 20,656
- Source: Kaggle Formula 1 telemetry dataset

Although many telemetry features were available, this study focused on variables most closely related to driver behavior and fuel efficiency.

3. Feature Selection and Target Variable

Selected Input Features

- Speed:** Higher speeds increase aerodynamic drag and fuel usage.
- Gear Number:** Higher gears generally allow lower RPM at the same speed, improving efficiency.
- Throttle Position:** Represents engine load; aggressive throttle use increases fuel consumption.

Fuel Efficiency Score

Because the dataset does not provide direct fuel consumption data, a **Fuel Efficiency Score** was calculated using speed, RPM, and throttle values. Higher scores indicate more efficient driving, where speed is maintained with lower engine load.

4. Data Preprocessing

Before training, the dataset was prepared using the following steps:

- Removal of invalid values (NaN or Infinity)
- Min-max normalization of inputs and outputs
- Random shuffling of samples
- Splitting the dataset into:
 - 80% training data
 - 20% testing data

These steps ensure stable training and fair model evaluation.

5. Neural Network Architecture

The model is a feedforward neural network designed for regression.

Architecture Overview

- Input layer: 3 neurons (Speed, Gear, Throttle)
- Hidden layer 1: 10 neurons with ReLU activation
- Hidden layer 2: 8 neurons with ReLU activation
- Output layer: 1 neuron with linear activation

This structure provides enough capacity to learn non-linear relationships while avoiding overfitting.

6. Training Configuration

The network was trained using stochastic gradient descent with the following settings:

- Learning rate: 0.01
- Epochs: 100
- Batch size: 8
- Loss function: Mean Squared Error (MSE)

During training, the loss consistently decreased, indicating that the model was learning meaningful patterns from the data.

7. Results and Analysis

Observed Trends

- Moderate speeds tend to be more fuel-efficient than extremely high speeds.
- Using higher gears at appropriate speeds improves efficiency.
- Smooth, moderate throttle usage results in better fuel efficiency than aggressive acceleration.

Model Capabilities

- Predicts fuel efficiency from driving inputs
- Identifies efficient driving behavior patterns
- Can support eco-driving recommendations or driver training systems

Limitations

- Fuel efficiency is estimated, not directly measured
 - Data is from a racing environment
 - External factors such as road conditions and weather are not included
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8. Applications

Potential applications of this model include:

- Eco-driving assistance systems
 - Driver training and performance analysis
 - Autonomous vehicle efficiency planning
 - Motorsport strategy optimization
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9. Conclusion

This case study demonstrates that neural networks can effectively model the relationship between driving behavior and fuel efficiency. By analyzing real Formula 1 telemetry data, the model learns complex non-linear patterns and provides insights into how speed, gear selection, and throttle control affect fuel consumption.

Overall, the project shows that data-driven approaches can support more fuel-efficient driving strategies in both racing and real-world scenarios.